Chapter 13, Problem 95CP

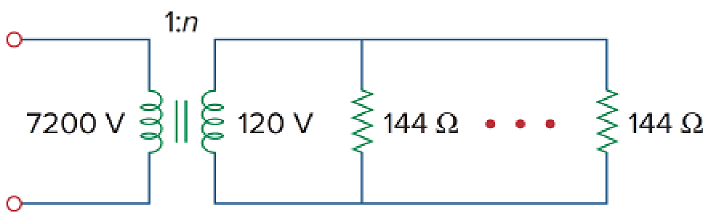
Problem

Ten bulbs in parallel are supplied by a 7,200/120-V transformer as shown in Fig. 13.147, where the bulbs are modeled by the 144-Ω resistors. Find:

(a) the turns ratio *n*,

(b) the current through the primary winding.

Figure 13.147



Step-by-step solution

Step 1 of 2

(A) $n = \frac{V_S}{V_P}$

$= \frac{120}{7200} = \frac{1}{60}$.

Step 2 of 2

(B) $I_S = 10 \times \frac{120}{144} = \frac{1200}{144}$

$S = V_P I_P = V_S I_S$

$I_P = \frac{V_S I_S}{V_P}$

$= \left(\frac{1}{60}\right) \times \frac{1200}{144}$

$I_P = 139 \ mA$.